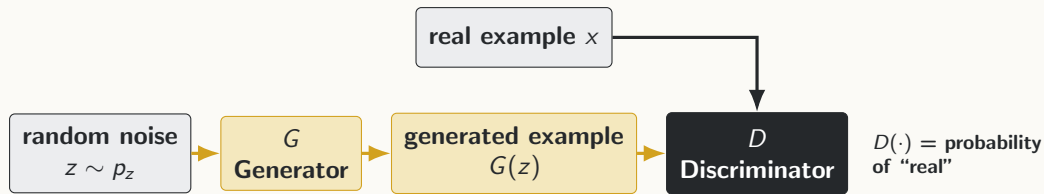


A GAN is one game with two opposite goals



THE TRAINING OBJECTIVE

$$\min_G \max_D V(D, G) = \underbrace{\mathbb{E}_{x \sim p_{\text{data}}} [\log D(x)]}_{D \text{ recognizes real data}} + \underbrace{\mathbb{E}_{z \sim p_z} [\log(1 - D(G(z)))]}_{D \text{ rejects fakes; } G \text{ fools } D}$$

D tries to classify correctly.

G tries to make its fakes look real to D .

D gives G feedback, not an answer key

WHAT D MAXIMIZES

$$\log D(x)$$

Reward D for calling real data real.

$D(x)$ moves toward 1.

$$\log(1 - D(G(z)))$$

Reward D for rejecting generated data.

$D(G(z))$ moves toward 0.

G can change only the fake path.
It pushes $D(G(z))$ toward 1.

HOW G LEARNS

1

Create

Noise becomes a fake.

2

Judge

D scores the fake.

3

Backpropagate

Gradient flows through D .

4

Adjust

G changes and tries again.

Success means D can no longer tell the difference

THE IDEAL BALANCE

$$p_g = p_{\text{data}}$$

The generated distribution matches the real data distribution.

$$D_G^*(x) = \frac{1}{2}$$

WHAT WE USE IN PRACTICE

$$L_G = -\log D(G(z))$$

Same destination. A stronger early gradient when G is losing.

What $D = \frac{1}{2}$ means

Real and generated examples are indistinguishable, so D can do no better than guess.

A confused discriminator is success only when G learned the distribution; not merely because D failed.